

DPEN022: Enabling Mathematics B

Class Questions Week 2

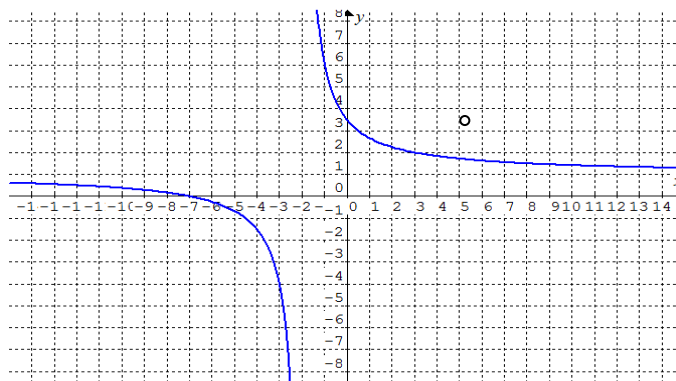
Week 2A

Trigonometry - Trigonometric Identities

1. Show that $\theta = 60^\circ$ satisfies the identity $\sin^2 \theta + \cos^2 \theta = 1$.
2. (i) If θ is an acute angle and $\sin \theta = \frac{2}{\sqrt{5}}$, use a Pythagorean identity to find $\cos \theta$ and $\tan \theta$.
(ii) If β is an obtuse angle and $\cos \beta = -\frac{2}{3}$ use a Pythagorean identity to find $\sin \beta$ and $\tan \beta$.
(iii) If $\tan A = \frac{1}{3}$ and A is a reflex angle, use a Pythagorean identity to find $\sec A$ and $\cos A$.
3. Find all values for $\cos \theta$ given $\sin \theta = \frac{-1}{5}$.
4. Simply the following:
(i) $2 - 2\cos^2 \theta$ (ii) $\sec^2 \theta (1 - \cos^2 \theta)$ (iii) $\tan \theta \cos \theta$
(iv) $1 - \sin^2 \theta - \cos^2 \theta - 1$ (v) $\frac{\tan \theta \sec \theta}{\sec^2 \theta}$ (vi) $\tan \theta \sqrt{1 - \sin^2 \theta}$
(vii) $\frac{\sin \theta}{\cos \theta} + \frac{\cos \theta}{\sin \theta}$
5. Prove the following identities:
(i) $\frac{2\cos^2 \theta}{1 - \sin \theta} = 2 + 2\sin \theta$ (ii) $(\sec^2 \theta - 1)\cos^2 \theta = \sin^2 \theta$
(iii) $\frac{\sin \theta}{1 + \cos \theta} = \frac{1 - \cos \theta}{\sin \theta}$ (iv) $\sin \theta \tan \theta = \frac{1 - \cos^2 \theta}{\cos \theta}$
6. Solve the following equations for $0^\circ \leq x \leq 360^\circ$ the following identities:
(i) $2\sin^2 x = \cos x + 2$ (ii) $3 - 3\sin x = \cos^2 x$ (iii) $\tan^2 x + 2\sec^2 x - 3 = 0$

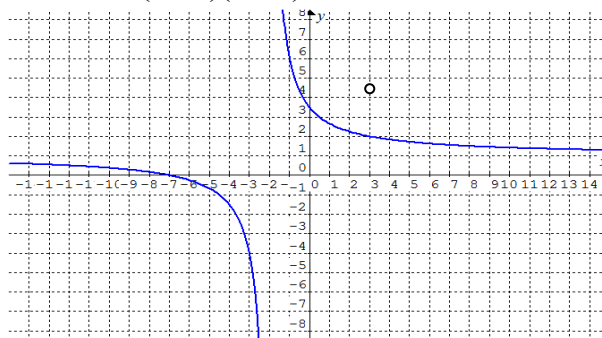
Week 2D

1. Use the graph below to find the following limits if they exist:



- (i) $\lim_{x \rightarrow 3} f(x)$ (ii) $\lim_{x \rightarrow -2} f(x)$
 (iii) $\lim_{x \rightarrow \infty} f(x)$ (iv) $\lim_{x \rightarrow -7} f(x)$

2. Consider the function $y = \frac{x^2 + 4x - 21}{(x - 3)(x + 2)}$ drawn below.



- (i) For what two values of x is the function undefined?
 (ii) Write the equation of the vertical asymptote on the graph.
 (iii) Write the equation of the horizontal asymptote on the graph.
3. Evaluate the following limits if they exist:

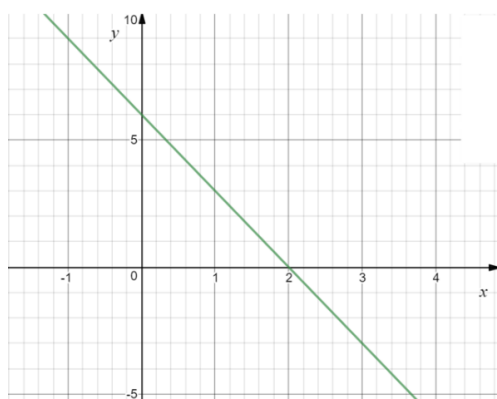
- (i) $\lim_{x \rightarrow 2} 5x - 3$ (ii) $\lim_{x \rightarrow 1} \frac{x^2 - 9}{x^2 + 1}$
 (iii) $\lim_{x \rightarrow 3} \frac{3 - x}{x}$ (iv) $\lim_{x \rightarrow 1} 3^{2x-1}$
 (v) $\lim_{x \rightarrow 0} e^x$ (vi) $\lim_{x \rightarrow 3} \frac{1}{2^{2x-1}} - 5$
 (vii) $\lim_{x \rightarrow 3} \log_4(3x - 5)$ (viii) $\lim_{x \rightarrow 2} \frac{\ln(x-1)}{(x-1)}$
 (ix) $\lim_{x \rightarrow \frac{\pi}{3}} \tan x$ (x) $\lim_{x \rightarrow \frac{\pi}{6}} \cot 4x$
 (xi) $\lim_{x \rightarrow 4} \frac{x^2 - 16}{x - 4}$ (xii) $\lim_{x \rightarrow 1} \frac{x^2 - 3x + 2}{x - 1}$
 (xiii) $\lim_{x \rightarrow 2} \frac{x^2 + x - 6}{x^2 - 4}$

4. By analysing each graph below, find the following limits, if they exist:

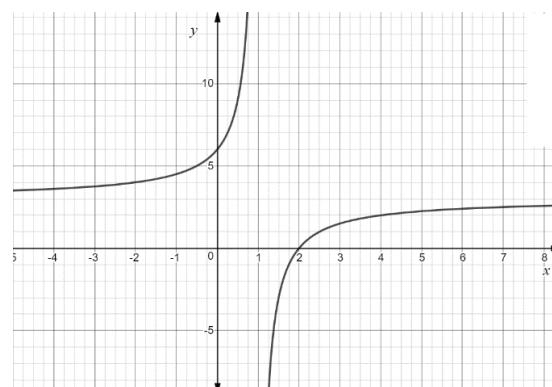
(i) $\lim_{x \rightarrow \infty} f(x)$

(ii) $\lim_{x \rightarrow -\infty} f(x)$

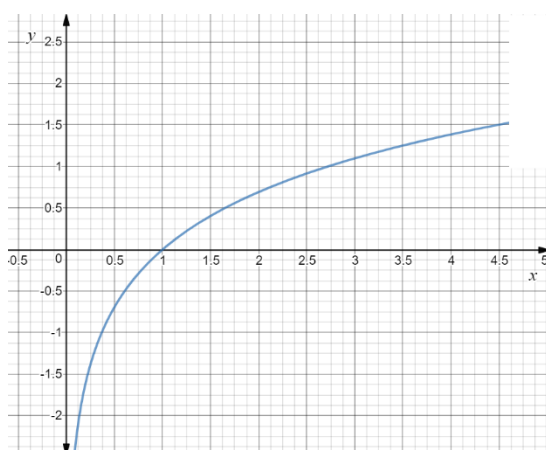
Graph 1



Graph 2



Graph 3



5. Evaluate the following limits if they exist:

(i) $\lim_{x \rightarrow \infty} \frac{x^2 + x - 6}{2x^2 - 4}$

(ii) $\lim_{x \rightarrow \infty} \frac{x - 6}{x^2 - 4}$

(iii) $\lim_{x \rightarrow \infty} \frac{4x^3 + x - 8}{2x^5 - 4x + 2}$

(iv) $\lim_{x \rightarrow \infty} \frac{7x^2 + x - 6}{2x^2 - 4}$